

Practical Strategies Leveraging AI and Digital Twins for the Design and Management of Future Optical Networks

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Abstract: This paper explores operationally reliable combinations of AI and physics through digital twins to optimize optical networks. Key methodologies and applications are discussed, addressing security, scalability, automation and sustainability.

1. Introduction

Contemporary optical networks face multifaceted challenges and opportunities driven by geopolitical instability, climate change, emerging quantum technologies, and artificial intelligence (AI). Notably, agentic AI accelerates capacity demands while optical networks approach various physical limitations. The consequential pressure on costs is driving the market towards scalable network architectures leveraging pluggable components, low-complexity amplifiers, and disaggregated infrastructure. However, this is also pushing further complexity to the control plane, necessitating practical solutions that ensure optical networks continue meeting capacity demands while addressing escalating requirements for security, reliability, sustainability, and cost-effectiveness.

Over the past decade, optical network research has significantly evolved, transitioning from passive optical performance monitoring (OPM) and physics-based quality of transmission (QoT) models to margin optimization and automated failure detection [1]. Introduced later, machine learning (ML) [2] and digital twins (DT) [3, 4] reinforced the data-driven network paradigm. Following in from that evolution, a broad consensus recently emerged on promising agentic approaches combining large language models (LLM), ML, DT and physics models [5, 6] with an unchanged goal: leverage monitoring and automation to optimize all network aspects, which is much aligned with current and future needs. Yet, as proposed control structures complexify, their implementation may be perceived by stakeholders as increasingly unrealistic in a context where some or most of the simplest features have yet to be deployed, and trust is a major issue. Thus, this paper proposes a practical approach to the harmonious integration of agentic AI within optical networks, using physics to keep the AI in check. We later focus on the key features required of the optical network digital twin (ONDT) for effective network optimization.

2. LLM and DT co-existence in the agentic design and management of future optical networks

QoT estimation and automated failure analysis are the two key points to address for an optimized physical layer. While QoT accuracy directly impacts the effective use of network resources through margins, the latter directly impacts the availability hence the quality of service (QoS) through the reduction of the mean-time-to-repair (MTTR). For both applications, a combination of LLMs with hybrid ML-physics methods would be ideal, in particular to alleviate the risk of LLM hallucinations while greatly simplifying the integration of many high-level, telemetry-related applications that would have required massive development efforts otherwise.

In Fig. 1a, we represent the integration of an AI agent and ONDT with a standard network management suite (NMS) controlling the network's optical layer as a possible synthesis of the current literature. The AI agent contains a central LLM that interacts with the user, with RAGs trained on technical literature to provide both knowledge and coding references, and with two adversarial LLMs checking and eventually correcting actions proposed by the central LLM: one responsible for the coherence with physics (LLM_b), the other for the consistency of the proposed code (LLM_c). Both communicate with specific blocks of the ONDT: P-DT emulates the physics - e.g. for QoT estimation - while SW-DT emulates the control software, thus guaranteeing the AI-generated control code will properly execute and achieve the expected outcome in practice. Importantly, existing NMS - ultimately interacting with the physical layer - would only require minimal modifications to be compatible with such agent.

Let us illustrate the workflow triggered by the generic user request "Do X" (1). LLM_a first calls the knowledge RAG, essentially asking: "How to do X?" (2) to which it answers "By doing Y" (3), Y being a set of actions known

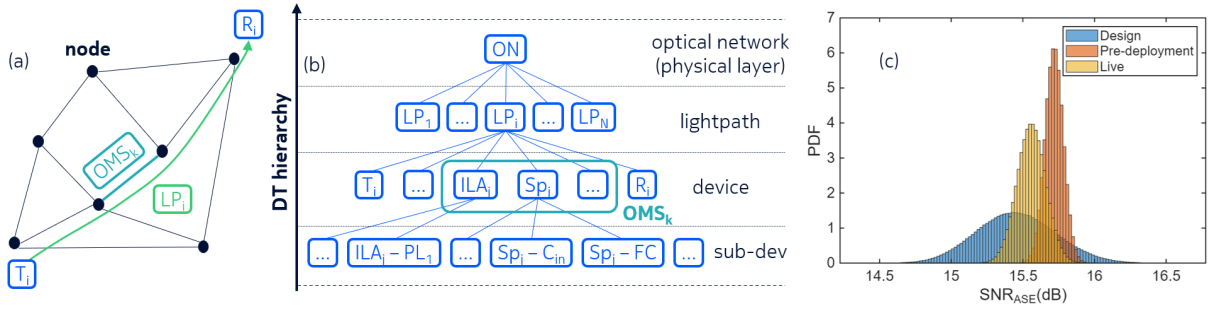


Fig. 2. (a) Schematic representation of an ON's physical layer and (c) tree diagram for ONDT hierarchy and dependencies

looking into safer options. In other words, there cannot be any robust automation in the AI-DT architecture without a proper statistical estimation of the DT's accuracy. In the following, we further focus on how to automatically estimate DT's accuracy in the QoT estimation use case.

4. Estimating the accuracy of an ONDT's output: the lightpath QoT use case

The impact of gaussian-distributed input parameter error on QoT estimation accuracy has been extensively studied in [7]. Yet, for ONDTs, it's important to consider that input errors are often - in practice - not gaussian, and strongly depend on the ON's status at any given time. At design, accuracy is typically low, QoT inputs being retrieved from datasheet and calibration values that merely represent typical or worst-case devices. Pre-deployment, individual device calibration guarantees much better QoT estimation for the future ON. Yet, once the ON is live, the accuracy of QoT estimation made from prior knowledge progressively decreases as the ON evolves and ages [8], in addition to the significant QoT fluctuations due to polarization-dependent loss (PDL) [9]. Then, only up-to-date monitoring data can maintain if not improve QoT accuracy [10]. Here, each ON will have a unique set of prior knowledge and monitoring tools, making the systematic evaluation of the QoT prediction's accuracy particularly challenging. This however can be achieved through Monte-Carlo simulations with any QoT estimation tool.

We illustrate it by evaluating the QoT estimation error for a typical lightpath from EDFAs noise figure (NF) values. For simplicity, we only consider ASE noise. The datasheet states that the NFs of the ten EDFAs will be between 6 and 9dB, which translates to NF values uniformly distributed in that interval. Pre-deployment, each EDFA's NF is calibrated with a method producing a zero-mean, normal error with a 0.2dB-standard-deviation (std). Live, the QoT is directly measured but fluctuates over time due to PDL. These fluctuations are normal with 0.1dB-std. These three stages are represented by the PDFs of SNR_{ase} in Fig.2c, each obtained from 10⁶ Monte-Carlo trials. The worst QoT estimation is at design when only datasheet values are known. Thanks to individual EDFA calibration pre-deployment, the accuracy drastically improves. Yet, the residual offset between best prediction and live QoT PDF is due to the limited accuracy of the calibration and live QoT fluctuations, which should ideally be included at each evaluation stage to properly evaluate the actual risk of a QoT below FEC limit.

5. Conclusion

We explored operationally reliable combinations of agentic AI and physics through digital twins to optimize key aspects of optical networks. We particularly focused on the physical layer DT, highlighting key features such as the Monte-Carlo estimation of QoT estimation accuracy for robust automation through the AI agent.

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